SOME IMPORTANT CHARACTERISTICS OF DIFFERENT Li-Ion CELL CHEMISTRIES

Chemistry	Cell Voltage (In volts)	Energy Density (WH/kg)	Self-discharge (% / month)	Temperature range (in °C)	Round trip efficiency (in %)	Cycles Times (maximum)	Cost	Weight	Memory	Safety
LCO	3.7	150-180	10	-40 to 80	90	1000	Expensive	Light	No	Bad
LMO	3.7	100-150	1	-40 to 85	95	700	Cheap	Heavy	No	Good
LFP	3.2	90-160	5	-40 to 80	95	12,000	Cheap	Light	No	Very good
NMC	3.6	150-220	10	-40 to 70	93	2000	Expensive	Light	No	Bad
NCA	3.6	200-260			90	3000	Expensive	Light	No	Good
LTO	2.4	50-80	5	-40 to 55	97	20,000	Expensive	Light	No	Very good

plications in low-powered EVs, such as e-rikshaws and e-bikes. These are also suitable for industrial equipment that have previously relied on lead-acid batteries, such as forklifts, heavy machines, and other equipment.

NMC: High-performing, affordable cell

Lithium nickel manganese cobalt oxide (NMC) cells have a cathode made up of a combination of nickel, manganese, and cobalt—commonly in the ratio of 60%, 20%, and 20%, respectively. The properties of the cell can be altered by changing the proportion of each element to achieve either a higher specific energy density or a higher specific power. The nominal voltage of the NMC cell is 3.7V and a high energy density of 150-220Wh/kg. These batteries have a low self-heating rate and are cheaper than most other chemistries, which makes them highly popular.

Application. NMC batteries are used in power tools and are one of the most preferred battery chemistries for EVs.

NCA: High energy and lasting cell

Lithium nickel cobalt aluminium oxide (NCA) cells are basically LCO cells with nickel and aluminium added as extra elements in their cathode. NCA cells have a high energy density of 200 to 260Wh/kg and a nominal voltage of 3.6V. The cells have a decent lifespan of 2000 charge cycles. Although the high proportion of nickel in the cathode improves the energy density of the cell, these cells are not stable and therefore require additional safety requirements to monitor their behaviour and keep users safe. NCA batteries can sustain a high charging current, thus enabling fast charging, but the limited availability of cobalt and nickel makes these batteries expensive.

Application. The NCA batteries have a high energy density, making them suitable for use in electric powertrains, grid energy storage, etc.

LTO: Long-lasting but low energy density cell

Lithium titanate (LTO), also known as li-titanate, cells incorporate advanced nanotechnology in their anode. Unlike other cells, which employ graphite, LTO's anode is made up of lithium titanate, a highly porous material that has up to 33x more surface area per gram than carbon. The high surface area allows faster charge and discharge rates. The LTO cells have a poor energy density of around 50 to 80 Wh/kg and has a nominal voltage of 2.4V. The high surface area enables faster movement of electrons, which enables discharge rates exceeding 30C for short period of time. (The charge and discharge current of a battery is measured in C-rate. Most portable batteries are rated at 1C.) LTO is one of the safest chemistries available in the market. These batteries have a very high operation life of 7,000 to 20,000 cycles.

Application. LTO batteries have been employed in the Japanese-only variant of Mitsubishi's i-MiEV and in some Honda bikes and cars. The LTO cells are not commonly used in consumer electronics due to their lower energy density, but their fast charging and discharging rate make them suitable for applications in heavy machinery, power tools, EVs, etc. In future, they may find application in storing wind and solar energy and creating smart grids.

Comparison

Li-ion cells have very different characteristics based on their internal construction. The table above compares some important characteristics of different Li-ion cell chemistries for reference.

Li-ion cell is one of the most important energy storage devices in today's time. The cell has a lot of different chemistries, and the properties of these cells depend on internal chemistry of the cells. Every single chemistry has its own pros and cons. Therefore, selection of the correct cell is crucial for achieving the best cost, performance, and weight of a product. EFY

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